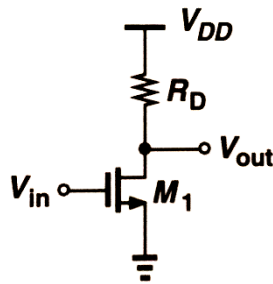


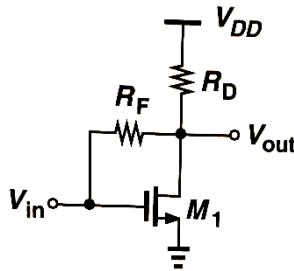
Due 09/29/2018 (Problem 1-9)
Due 10/2/2018 (Simulation problem)

Each problem from 1 to 9 carries 10 points.

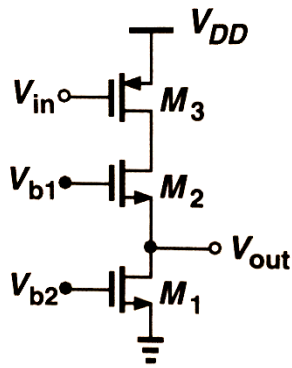
1. Suppose the common source stage of the figure below is to provide an output swing from 1V to 2.5V. Assuming $(W/L) = 50/0.5$, $R_D = 2k\Omega$, and $\lambda = 0$.
 - a. Find the input voltages that yield $V_{out} = 1V$ and $V_{out} = 2.5V$.
 - b. Find the drain current and transconductance of M_1 for both cases.
 - c. How much does the small signal gain, $g_m \times R_D$, vary as the output goes from 1V to 2.5V? (variation of small signal gain can be viewed as nonlinearity).



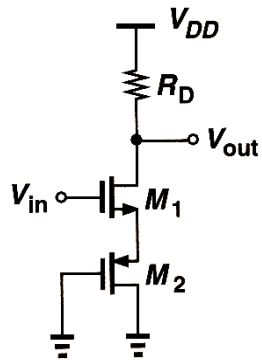
2. Sketch V_{out} vs. V_{in} for the following circuit as V_{in} varies from 0 to V_{dd} . Find the important transition points.



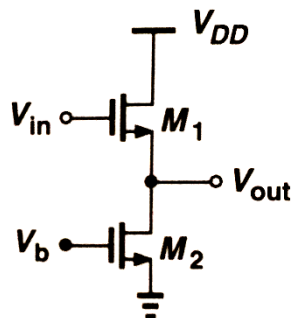
3. Sketch V_{out} vs. V_{in} for the following circuit as V_{in} varies from 0 to V_{dd} . Find the important transition points.



4. Assuming $\text{Lambd}a = \gamma = 0$, find the small signal gain of the circuit below.



5. A source follower can operate as a level shifter. Suppose the following circuit is designed to shift the voltage level by 1V, i.e. $V_{in} - V_{out} = 1V$.
- Find the dimensions of M1 and M2 if $I_{d1} = I_{d2} = 0.5\text{mA}$, $V_{gs2} - V_{gs1} = 0.5V$, and $\text{Lambd}a = \gamma = 0$.
 - Repeat part a. if $\gamma = 0.45\text{ V}^{-1}$ and $V_{in} = 2.5V$. What is the minimum input voltage for M2 to remain saturated?



- Problem 2.5 (a)
- Problem 2.5(b)
- Problem 2.5(e)
- Problem 3.15(c)

Simulation Problem (30 points):

Considering load resistor $R_D=500\Omega$, the capacitor C of 1pF , and $V_{DD}=1.3\text{V}$, you need to design a 2-stage cascaded common-source amplifier with a source follower driving 50Ω .

You are to use an 130nm CMOS process. The design goals for this circuit are as follows:

Low frequency voltage gain $> 10\text{dB}$

Maximum output swing $> 0.4\text{V}_{\text{pk-pk-se}}$ (peak-to-peak-single-ended)

Unity gain bandwidth $> 300\text{MHz}$

What you need to turn in for this problem are:

- (15 pts) Hand calculation results including the values of DC bias voltage and current, transistor size, small-signal gain, output resistance, and maximum output swing.
- (15 pts) Cadence simulation results including a plot of small-signal gain and phase, and information showing DC conditions of the MOSFET.

Report: Attach schematics with DC operating points, AC gain and phase, transient simulation showing results at the bias points with the output and input swings. Note that you have to decide the right bias points to allow all the transistors $M1$, $M2$ and $M3$ to be in saturation. You also have to ensure that none of these transistors go out of saturation at the maximum or the minimum input voltages.

